■ EXAMPLE 9.2

Consider the same two state system as described in Example 9.1. Suppose we are trying to estimate the state of the system with a fixed time lag. The discretization time step $T = 0.1$ and the standard deviation of the acceleration noise is 10. Figure 9.8 shows the percent improvement in state estimation that is available with fixed-lag smoothing. The figure shows percent improvement as a function of lag size, and for two different values of measurement noise. The values on the plot are based on the theoretical estimation-error covariance. As expected, the improvement in estimation accuracy is more dramatic as the measurement noise decreases. This was discussed at the end of Section 9.2.

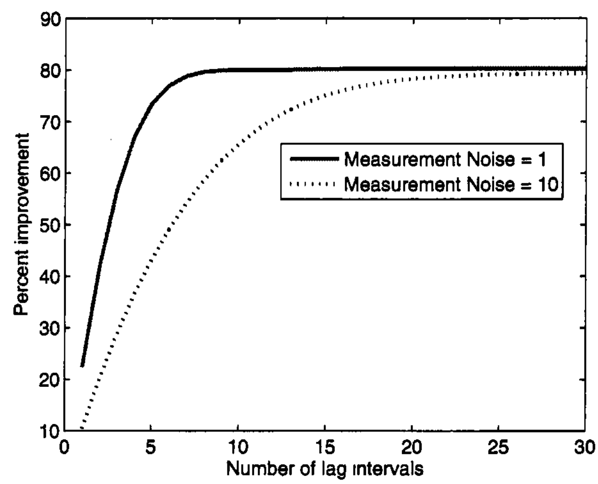

Figure 9.8 This shows the percent improvement of the trace of the estimation-error covariance of the smoothed estimate of the state (relative to the standard Kalman filter) for Example 9.2. As the number of lag intervals increases, the estimation error of the smoother decreases and the percent improvement increases. Also, as the measurement noise decreases, the improvement due to smoothing is more dramatic.

▽▽▽

9.4 FIXED-INTERVAL SMOOTHING

Suppose we have measurements for a fixed time interval. In fixed-interval smoothing we seek an estimate of the state at some of the interior points of the time interval. During the smoothing process we do not obtain any new measurements. Section 9.4.1 discusses the forward–backward approach to smoothing, which is perhaps the most straightforward smoothing algorithm. Section 9.4.2 discusses the RTS smoother, which is conceptually more difficult but is computationally cheaper than forward–backward smoothing.

9.4.1 Forward–backward smoothing

Suppose we want to estimate the state x_m based on measurements from $k = 1$ to $k = N$, where $N > m$. The forward–backward approach to smoothing obtains two estimates of x_m. The first estimate, $\hat{x}_f$, is based on the standard Kalman filter that operates from $k = 1$ to $k = m$. The second estimate, $\hat{x}_b$, is based on a Kalman filter that runs backward in time from $k = N$ back to $k = m$. The forward–backward approach to smoothing combines the two estimates to form an optimal smoothed estimate. This approach was first suggested in [Fra69].

Suppose that we combine a forward estimate $\hat{x}_f$ of the state and a backward estimate $\hat{x}_b$ of the state to get a smoothed estimate of x as follows:

$$\hat{x} = K_f\hat{x}_f + K_b\hat{x}_b \tag{9.55}$$

where K_f and K_b are constant matrix coefficients to be determined. Note that $\hat{x}_f$ and $\hat{x}_b$ are both unbiased since they are both outputs from Kalman filters. Therefore, if $\hat{x}$ is to be unbiased, we require $K_f + K_b = I$ (see Problem 9.9). This gives

$$\hat{x} = K_f\hat{x}_f + (I - K_f)\hat{x}_b \tag{9.56}$$

The covariance of the estimate can then be found as

$$\begin{aligned} P &= E[(x-\hat{x})(x-\hat{x})^T] \\ &= E\left\{[x - K_f\hat{x}_f - (I-K_f)\hat{x}_b][\cdots]^T\right\} \\ &= E\left\{[K_f(e_f - e_b) + e_b][\cdots]^T\right\} \\ &= E\left\{K_f(e_fe_f^T + e_be_b^T)K_f^T + e_be_b^T - K_fe_be_b^T - e_be_b^TK_f^T\right\} \end{aligned} \tag{9.57}$$

where $e_f = x - x_f$, $e_b = x - x_b$, and we have used the fact that $E(e_fe_b^T) = 0$. The estimates $\hat{x}_f$ and $\hat{x}_b$ are both unbiased, and e_f and e_b are independent (since they depend on separate sets of measurements). We can minimize the trace of P with respect to K_f using results from Equation (1.66) and Problem 1.4:

$$\begin{aligned} \frac{\partial \mathrm{Tr}(P)}{\partial K_f} &= 2E\left\{K_f(e_fe_f^T + e_be_b^T) - e_be_b^T\right\} \\ &= 2[K_f(P_f + P_b) - P_b] \end{aligned} \tag{9.58}$$

where $P_f = E(e_fe_f^T)$ is the covariance of the forward estimate, and $P_b = E(e_be_b^T)$ is the covariance of the backward estimate. Setting this equal to zero to find the optimal value of K_f gives

$$\begin{aligned} K_f &= P_b(P_f + P_b)^{-1} \\ K_b &= P_f(P_f + P_b)^{-1} \end{aligned} \tag{9.59}$$

The inverse of $(P_f + P_b)$ always exists since both covariance matrices are positive definite. We can substitute this result into Equation (9.57) to find the covariance of the fixed-interval smoother as follows:

$$\begin{aligned} P = {} & P_b(P_f+P_b)^{-1}(P_f+P_b)(P_f+P_b)^{-1}P_b + \\ & P_b - P_b(P_f+P_b)^{-1}P_b - P_b(P_f+P_b)^{-1}P_b \end{aligned} \tag{9.60}$$

Using the identity $(A+B)^{-1} = B^{-1}(AB^{-1}+I)^{-1}$ (see Problem 9.2), we can write the above equation as

$$\begin{aligned} P &= (P_fP_b^{-1}+I)^{-1}(P_f+P_b)(P_b^{-1}P_f+I)^{-1}+ \\ &\quad P_b-(P_fP_b^{-1}+I)^{-1}P_b-(P_fP_b^{-1}+I)^{-1}P_b \end{aligned} \tag{9.61}$$

Multiplying out the first term, and again using the identity $(A+B)^{-1} = B^{-1}(AB^{-1}+I)^{-1}$ on the last two terms, results in

$$\begin{aligned} P &= \left[(P_b^{-1}+P_f^{-1})^{-1}+(P_b^{-1}P_fP_b^{-1}+P_b^{-1})^{-1}\right](P_b^{-1}P_f+I)^{-1}+ \\ &\quad P_b-2(P_b^{-1}P_fP_b^{-1}+P_b^{-1})^{-1} \end{aligned} \tag{9.62}$$

From the matrix inversion lemma of Equation (1.39) we see that

$$(P_b^{-1}P_fP_b^{-1}+P_b^{-1})^{-1} = P_b-(P_f^{-1}+P_b^{-1})^{-1} \tag{9.63}$$

Substituting this into Equation (9.62) gives

$$\begin{aligned} P &= P_b(P_b^{-1}P_f+I)^{-1}+P_b-2P_b+2(P_f^{-1}+P_b^{-1})^{-1} \\ &= (P_b^{-1}P_fP_b^{-1}+P_b^{-1})^{-1}-P_b+2(P_f^{-1}+P_b^{-1})^{-1} \\ &= P_b-(P_f^{-1}+P_b^{-1})^{-1}-P_b+2(P_f^{-1}+P_b^{-1})^{-1} \\ &= (P_f^{-1}+P_b^{-1})^{-1} \end{aligned} \tag{9.64}$$

These results form the basis for the fixed-interval smoothing problem. The system model is given as

$$\begin{aligned} x_k &= F_{k-1}x_{k-1}+w_{k-1} \\ y_k &= H_kx_k+v_k \\ w_k &\sim (0,Q_k) \\ v_k &\sim (0,R_k) \end{aligned} \tag{9.65}$$

Suppose we want a smoothed estimate at time index m. First we run the forward Kalman filter normally, using measurements up to and including time m.

1. Initialize the forward filter as follows:

$$\begin{aligned} \hat{x}_{f0}^+ &= E(x_0) \\ P_{f0}^+ &= E\left[(x_0-\hat{x}_{f0}^+)(x_0-\hat{x}_{f0}^+)^T\right] \end{aligned} \tag{9.66}$$

2. For $k = 1, \cdots, m$, perform the following:

$$\begin{aligned} P_{fk}^- &= F_{k-1}P_{f,k-1}^+F_{k-1}^T+Q_{k-1} \\ K_{fk} &= P_{fk}^-H_k^T(H_kP_{fk}^-H_k^T+R_k)^{-1} \\ &= P_{fk}^+H_k^TR_k^{-1} \\ \hat{x}_{fk}^- &= F_{k-1}\hat{x}_{f,k-1}^+ \\ \hat{x}_{fk}^+ &= \hat{x}_{fk}^-+K_{fk}(y_k-H_k\hat{x}_{fk}^-) \\ P_{fk}^+ &= (I-K_{fk}H_k)P_{fk}^- \end{aligned} \tag{9.67}$$

At this point we have a forward estimate for x_m, along with its covariance. These quantities are obtained using measurements up to and including time m.

The backward filter needs to run backward in time, starting at the final time index N. Since the forward and backward estimates must be independent, none of the information that was used in the forward filter is allowed to be used in the backward filter. Therefore, P_{bN}^- must be infinite:

$$P_{bN}^- = \infty \tag{9.68}$$

We are using the minus superscript on P_{bN}^- to indicate the backward covariance at time N before the measurement at time N is processed. (Recall that the filtering is performed backward in time.) So P_{bN}^- will be updated to obtain P_{bN}^+ after the measurement at time N is processed. Then it will be extrapolated backward in time to obtain $P_{b,N-1}^-$, and so on.

Now the question arises how to initialize the backward state estimate $\hat{x}_{bk}^-$ at the final time $k = N$. We can solve this problem by introducing the new variable

$$s_k = P_{bk}^{-1}\hat{x}_{bk} \tag{9.69}$$

A minus or plus superscript can be added on all the quantities in the above equation to indicate values before or after the measurement at time k is taken into account. Since $P_{bN}^- = \infty$ it follows that

$$s_N^- = 0 \tag{9.70}$$

The infinite boundary condition on P_{bk}^- means that we cannot run the standard Kalman filter backward in time because we have to begin with an infinite covariance. Instead we run the information filter from Section 6.2 backward in time. This can be done by writing the system of Equation (9.65) as

$$\begin{aligned}
x_{k-1} &= F_{k-1}^{-1}x_k + F_{k-1}^{-1}w_{k-1} \\
&= F_{k-1}^{-1}x_k + w_{b,k-1} \\
y_k &= H_k x_k + v_k \\
w_{bk} &\sim (0, F_k^{-1}Q_k F_k^{-T}) \\
v_k &\sim (0, R_k)
\end{aligned} \tag{9.71}$$

Note that F_k^{-1} should always exist if it comes from a real system, because F_k comes from a matrix exponential that is always invertible (see Sections 1.2 and 1.4). The backward information filter can be written as follows.

1. Initialize the filter with $\mathcal{I}_{bN}^- = 0$.
2. For $k = N, N-1, \cdots$, perform the following:

$$\begin{aligned}
\mathcal{I}_{bk}^+ &= \mathcal{I}_{bk}^- + H_k^T R_k^{-1} H_k \\
K_{bk} &= (\mathcal{I}_{bk}^+)^{-1} H_k^T R_k^{-1} \\
\hat{x}_{bk}^+ &= \hat{x}_{bk}^- + K_{bk}(y_k - H_k \hat{x}_{bk}^-) \\
\mathcal{I}_{b,k-1}^- &= \left[F_{k-1}^{-1}(\mathcal{I}_{bk}^+)^{-1}F_{k-1}^{-T} + F_{k-1}^{-1}Q_{k-1}F_{k-1}^{-T}\right]^{-1} \\
&= F_{k-1}^T\left[(\mathcal{I}_{bk}^+)^{-1} + Q_{k-1}\right]^{-1}F_{k-1} \\
&= F_{k-1}^T\left[Q_{k-1}^{-1} - Q_{k-1}^{-1}(\mathcal{I}_{bk}^+ + Q_{k-1}^{-1})^{-1}Q_{k-1}^{-1}\right]F_{k-1} \\
\hat{x}_{b,k-1}^- &= F_{k-1}^{-1}\hat{x}_{bk}^+
\end{aligned} \tag{9.72}$$

The first form for $\mathcal{I}_{b,k-1}^-$ above requires the inversion of $\mathcal{I}_{bk}^+$. Consider the first time step for the backward filter (i.e., at $k = N$). The information matrix $\mathcal{I}_{bN}^-$ is initialized to zero, and then the first time through the above loop we set $\mathcal{I}_{bk}^+ = \mathcal{I}_{bk}^- + H_k^T R_k^{-1} H_k$. If there are fewer measurements than states, $H_k^T R_k^{-1} H_k$ will always be singular and, therefore, $\mathcal{I}_{bk}^+$ will be singular at $k = N$. Therefore, the first form given above for $\mathcal{I}_{b,k-1}^-$ will not be computable. In practice we can get around this by initializing $\mathcal{I}_{bN}^-$ to a small nonzero matrix instead of zero.

The third form for $\mathcal{I}_{b,k-1}^-$ above has its own problems. It does not require the inversion of $\mathcal{I}_{bk}^+$, but it does require the inversion of Q_{k-1}. So the third form of $\mathcal{I}_{b,k-1}^-$ is not computable unless Q_{k-1} is nonsingular. Again, in practice we can get around this by making a small modification to Q_{k-1} so that it is numerically nonsingular.

Since we need to update $s_k = \mathcal{I}_{bk}\hat{x}_{bk}$ instead of $\hat{x}_{bk}$ (because of initialization issues) as defined in Equation (9.69), we rewrite the update equations for the state estimate as follows:

$$\begin{aligned}
\hat{x}_{bk}^+ &= \hat{x}_{bk}^- + K_{bk}(y_k - H_k \hat{x}_{bk}^-) \\
s_k^+ &= \mathcal{I}_{bk}^+ \hat{x}_{bk}^+ \\
&= \mathcal{I}_{bk}^+ \hat{x}_{bk}^- + \mathcal{I}_{bk}^+ K_{bk}(y_k - H_k \hat{x}_{bk}^-)
\end{aligned} \tag{9.73}$$

Now note from Equation (6.33) that we can write $\mathcal{I}_{bk}^+ = \mathcal{I}_{bk}^- + H_k^T R_k^{-1} H_k$, and $K_{bk} = P_{bk}^+ H_k^T R_k^{-1}$. Substituting these expressions into the above equation for s_k^+ gives

$$\begin{aligned}
s_k^+ &= \mathcal{I}_{bk}^- \hat{x}_{bk}^- + H_k^T R_k^{-1} H_k \hat{x}_{bk}^- + H_k^T R_k^{-1}(y_k - H_k \hat{x}_{bk}^-) \\
&= s_k^- + H_k^T R_k^{-1} y_k
\end{aligned} \tag{9.74}$$

We combine this with Equation (9.72) to write the backward information filter as follows.

1. Initialize the filter as follows:

$$\begin{aligned}
s_N^- &= 0 \\
\mathcal{I}_{bN}^- &= 0
\end{aligned} \tag{9.75}$$

2. For $k = N, N-1, \cdots, m+1$, perform the following:

$$\begin{aligned}
\mathcal{I}_{bk}^+ &= \mathcal{I}_{bk}^- + H_k^T R_k^{-1} H_k \\
s_k^+ &= s_k^- + H_k^T R_k^{-1} y_k \\
\mathcal{I}_{b,k-1}^- &= \left[F_{k-1}^{-1} (\mathcal{I}_{bk}^+)^{-1} F_{k-1}^{-T} + F_{k-1}^{-1} Q_{k-1} F_{k-1}^{-T}\right]^{-1} \\
&= F_{k-1}^T \left[(\mathcal{I}_{bk}^+)^{-1} + Q_{k-1}\right]^{-1} F_{k-1} \\
&= F_{k-1}^T \left[Q_{k-1}^{-1} - Q_{k-1}^{-1} (\mathcal{I}_{bk}^+ + Q_{k-1}^{-1})^{-1} Q_{k-1}^{-1}\right] F_{k-1} \\
s_{k-1}^- &= \mathcal{I}_{b,k-1}^- F_{k-1}^{-1} (\mathcal{I}_{bk}^+)^{-1} s_k^+
\end{aligned} \tag{9.76}$$

3. Perform one final time update to obtain the backward estimate of x_m:

$$
\begin{aligned}
\mathcal{I}_{bm}^- &= Q_m^{-1} - Q_m^{-1} F_m^{-1} (\mathcal{I}_{b,m+1}^+ + F_m^{-T} Q_m^{-1} F_m^{-1})^{-1} F_m^{-T} Q_m^{-1} \\
P_{bm}^- &= (\mathcal{I}_{bm}^-)^{-1} \\
s_m^- &= \mathcal{I}_{bm}^- F_m^{-1} (\mathcal{I}_{b,m+1}^+)^{-1} s_{m+1}^+ \\
\hat{x}_{bm}^- &= (\mathcal{I}_{bm}^-)^{-1} s_m^-
\end{aligned} \tag{9.77}
$$

Now we have the backward estimate $\hat{x}_{bm}^-$ and its covariance P_{bm}^-. These quantities are obtained from measurements $m+1, m+2, \cdots, N$.

After we obtain the backward quantities as outlined above, we combine them with the forward quantities from Equation (9.67) to obtain the final state estimate and covariance:

$$
\begin{aligned}
K_f &= P_{bm}^- (P_{fm}^+ + P_{bm}^-)^{-1} \\
\hat{x}_m &= K_f \hat{x}_{fm}^+ + (I - K_f) \hat{x}_{bm}^- \\
P_m &= \left[(P_{fm}^+)^{-1} + (P_{bm}^-)^{-1} \right]^{-1}
\end{aligned} \tag{9.78}
$$

We can obtain an alternative equation for $\hat{x}_m$ by manipulating the above equations. If we substitute for K_f in the above expression for $\hat{x}_m$ then we obtain

$$
\begin{aligned}
\hat{x}_m &= P_{bm}^- (P_{fm}^+ + P_{bm}^-)^{-1} \hat{x}_{fm}^+ + \left[I - P_{bm}^- (P_{fm}^+ + P_{bm}^-)^{-1} \right] \hat{x}_{bm}^- \\
&= P_{bm}^- (P_{fm}^+ + P_{bm}^-)^{-1} \hat{x}_{fm}^+ + \left[(P_{fm}^+ + P_{bm}^-) - P_{bm}^- \right] (P_{fm}^+ + P_{bm}^-)^{-1} \hat{x}_{bm}^- \\
&= P_{bm}^- (P_{fm}^+ + P_{bm}^-)^{-1} \hat{x}_{fm}^+ + P_{fm}^+ (P_{fm}^+ + P_{bm}^-)^{-1} \hat{x}_{bm}^-
\end{aligned} \tag{9.79}
$$

Using the matrix inversion lemma on the rightmost inverse in the above equation and performing some other manipulations gives

$$
\begin{aligned}
\hat{x}_m &= [(P_{bm}^- + P_{fm}^+) - P_{fm}^+](P_{fm}^+ + P_{bm}^-)^{-1} \hat{x}_{fm}^+ + \\
&\quad P_{fm}^+ \left[\mathcal{I}_{bm}^- - \mathcal{I}_{bm}^- (\mathcal{I}_{fm}^+ + \mathcal{I}_{bm}^-)^{-1} \mathcal{I}_{bm}^- \right] \hat{x}_{bm}^- \\
&= \left[I - P_{fm}^+ (P_{fm}^+ + P_{bm}^-)^{-1} \right] \hat{x}_{fm}^+ + \\
&\quad P_{fm}^+ \left[I - \mathcal{I}_{bm}^- (\mathcal{I}_{fm}^+ + \mathcal{I}_{bm}^-)^{-1} \right] \mathcal{I}_{bm}^- \hat{x}_{bm}^- \\
&= \left[I - P_{fm}^+ \mathcal{I}_{bm}^- (I + P_{fm}^+ \mathcal{I}_{bm}^-)^{-1} \right] \hat{x}_{fm}^+ + \\
&\quad P_{fm}^+ \left[I - \mathcal{I}_{bm}^- (I + P_{fm}^+ \mathcal{I}_{bm}^-)^{-1} P_{fm}^+ \right] \mathcal{I}_{bm}^- \hat{x}_{bm}^- \\
&= P_{fm}^+ \left[I - \mathcal{I}_{bm}^- (I + P_{fm}^+ \mathcal{I}_{bm}^-)^{-1} P_{fm}^+ \right] \mathcal{I}_{fm}^+ \hat{x}_{fm}^+ + \\
&\quad P_{fm}^+ \left[I - \mathcal{I}_{bm}^- (I + P_{fm}^+ \mathcal{I}_{bm}^-)^{-1} P_{fm}^+ \right] \mathcal{I}_{bm}^- \hat{x}_{bm}^-
\end{aligned} \tag{9.80}
$$

where we have relied on the identity $(A+B)^{-1} = B^{-1}(AB^{-1} + I)^{-1}$ (see Problem 9.2). The coefficients of $\hat{x}_{fm}^+$ and $\hat{x}_{bm}^-$ in the above equation both have a common factor which can be written as follows:

$$
\begin{aligned}
P_{fm}^{+}&\left[I - \mathcal{I}_{bm}^{-}(I + P_{fm}^{+}\mathcal{I}_{bm}^{-})^{-1}P_{fm}^{+}\right] \\
&= P_{fm}^{+} - P_{fm}^{+}\mathcal{I}_{bm}^{-}(I + P_{fm}^{+}\mathcal{I}_{bm}^{-})^{-1}P_{fm}^{+} \\
&= P_{fm}^{+} - P_{fm}^{+}(\mathcal{I}_{fm}^{+}P_{bm}^{-} + I)^{-1} \\
&= \left[P_{fm}^{+}(\mathcal{I}_{fm}^{+}P_{bm}^{-} + I) - P_{fm}^{+}\right](\mathcal{I}_{fm}^{+}P_{bm}^{-} + I)^{-1} \\
&= P_{bm}^{-}(\mathcal{I}_{fm}^{+}P_{bm}^{-} + I)^{-1} \\
&= (\mathcal{I}_{fm}^{+} + \mathcal{I}_{bm}^{-})^{-1}
\end{aligned}
\tag{9.81}
$$

Therefore, using Equation (9.78), we can write Equation (9.80) as

$$
\begin{aligned}
\hat{x}_m &= P_m\mathcal{I}_{fm}^{+}\hat{x}_{fm}^{+} + P_m\mathcal{I}_{bm}^{-}\hat{x}_{bm}^{-} \\
&= P_m\left(\mathcal{I}_{fm}^{+}\hat{x}_{fm}^{+} + \mathcal{I}_{bm}^{-}\hat{x}_{bm}^{-}\right)
\end{aligned}
\tag{9.82}
$$

Figure 9.9 illustrates how the forward–backward smoother works.

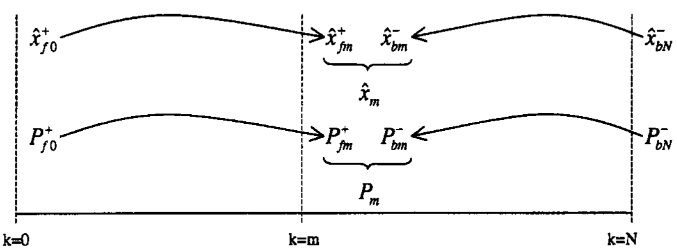

Figure 9.9 This figure illustrates the concept of the forward–backward smoother. The forward filter is run to obtain *a posteriori* estimates and covariances up to time m. Then the backward filter is run to obtain *a priori* estimates and covariances back to time m (i.e., *a priori* from a reversed time perspective). Then the forward and backward estimates and covariances at time m are combined to obtain the final estimate $\hat{x}_m$ and covariance P_m.

■ EXAMPLE 9.3

In this example we consider the same problem given in Example 9.1. Suppose that we want to estimate the position and velocity of the vehicle at $t = 5$ seconds. We have measurements every 0.1 seconds for a total of 10 seconds. The standard deviation of the measurement noise is 10, and the standard deviation of the acceleration noise is 10. Figure 9.10 shows the trace of the covariance of the estimation of the forward filter as it runs from $t = 0$ to $t = 5$, the backward filter as it runs from $t = 10$ back to $t = 5$, and the smoothed estimate at $t = 5$. The forward and backward filters both converge to the same steady-state value, even though the forward filter was initialized to a covariance of 20 for both the position and velocity estimation errors, and the backward filter was initialized to an infinite covariance. The smoothed filter has a covariance of about 7.6, which shows the dramatic improvement that can be obtained in estimation accuracy when smoothing is used.

$\nabla\nabla\nabla$

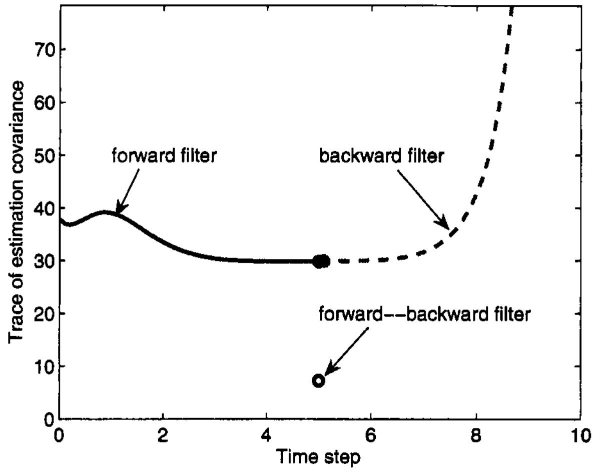

Figure 9.10 This shows the trace of the estimation-error covariance for Example 9.3. The forward filter runs from $t = 0$ to $t = 5$, the backward filter runs from $t = 10$ to $t = 5$, and the trace of the covariance of the smoothed estimate is shown at $t = 5$.

9.4.2 RTS smoothing

Several other forms of the fixed-interval smoother have been obtained. One of the most common is the smoother that was presented by Rauch, Tung, and Striebel, usually called the RTS smoother [Rau65]. The RTS smoother is more computationally efficient than the smoother presented in the previous section because we do not need to directly compute the backward estimate or covariance in order to get the smoothed estimate and covariance. In order to obtain the RTS smoother, we will first look at the smoothed covariance given in Equation (9.78) and obtain an equivalent expression that does not use P_{bm}. Then we will look at the smoothed estimate given in Equation (9.78), which uses the gain K_f, which depends on P_{bm}, and obtain an equivalent expression that does not use P_{bm} or $\hat{x}_{bm}$.

9.4.2.1 RTS covariance update First consider the smoothed covariance given in Equation (9.78). This can be written as

$$\begin{aligned} P_m &= \left[(P_{fm}^+)^{-1} + (P_{bm}^-)^{-1} \right]^{-1} \\ &= P_{fm}^+ - P_{fm}^+ (P_{fm}^+ + P_{bm}^-)^{-1} P_{fm}^+ \end{aligned} \tag{9.83}$$

where the second expression comes from an application of the matrix inversion lemma to the first expression (see Problem 9.3). From Equation (9.72) we see that

$$P_{bm}^- = F_m^{-1} \left[P_{b,m+1}^+ + Q_m \right] F_m^{-T} \tag{9.84}$$

Substituting this into the expression $(P_{fm}^+ + P_{bm}^-)^{-1}$ gives the following:

$$\begin{aligned}
(P_{fm}^+ + P_{bm}^-)^{-1} &= \left[P_{fm}^+ + F_m^{-1}(P_{b,m+1}^+ + Q_m)F_m^{-T}\right]^{-1} \\
&= \left[F_m^{-1}F_mP_{fm}^+F_m^TF_m^{-T} + F_m^{-1}(P_{b,m+1}^+ + Q_m)F_m^{-T}\right]^{-1} \\
&= \left[F_m^{-1}(F_mP_{fm}^+F_m^T + P_{b,m+1}^+ + Q_m)F_m^{-T}\right]^{-1} \\
&= F_m^T(F_mP_{fm}^+F_m^T + P_{b,m+1}^+ + Q_m)^{-1}F_m \\
&= F_m^T(P_{f,m+1}^- + P_{b,m+1}^+)^{-1}F_m
\end{aligned} \tag{9.85}$$

From Equations (6.26) and (9.76) recall that

$$\begin{aligned}
\mathcal{I}_{fm}^+ &= \mathcal{I}_{fm}^- + H_m^TR_m^{-1}H_m \\
\mathcal{I}_{bm}^+ &= \mathcal{I}_{bm}^- + H_m^TR_m^{-1}H_m
\end{aligned} \tag{9.86}$$

We can combine these two equations to obtain

$$\mathcal{I}_{b,m+1}^+ = \mathcal{I}_{b,m+1}^- + \mathcal{I}_{f,m+1}^+ - \mathcal{I}_{f,m+1}^- \tag{9.87}$$

Substituting this into Equation (9.78) gives

$$\begin{aligned}
P_{m+1} &= \left[\mathcal{I}_{f,m+1}^+ + \mathcal{I}_{b,m+1}^-\right]^{-1} \\
&= \left[\mathcal{I}_{b,m+1}^+ + \mathcal{I}_{f,m+1}^-\right]^{-1} \\
P_{m+1}^{-1} &= \mathcal{I}_{b,m+1}^+ + \mathcal{I}_{f,m+1}^- \\
P_{b,m+1}^+ &= \left[P_{m+1}^{-1} - \mathcal{I}_{f,m+1}^-\right]^{-1}
\end{aligned} \tag{9.88}$$

Substituting this into Equation (9.85) gives

$$\begin{aligned}
(P_{fm}^+ + P_{bm}^-)^{-1} &= F_m^T\left[P_{f,m+1}^- + \left(P_{m+1}^{-1} - \mathcal{I}_{f,m+1}^-\right)^{-1}\right]^{-1}F_m \\
&= F_m^T\mathcal{I}_{f,m+1}^-\left[\mathcal{I}_{f,m+1}^- + \mathcal{I}_{f,m+1}^-\left(P_{m+1}^{-1} - \mathcal{I}_{f,m+1}^-\right)^{-1}\mathcal{I}_{f,m+1}^-\right]^{-1}\mathcal{I}_{f,m+1}^-F_m \\
&= F_m^T\mathcal{I}_{f,m+1}^-\left(P_{f,m+1}^- - P_{m+1}\right)\mathcal{I}_{f,m+1}^-F_m
\end{aligned} \tag{9.89}$$

where the last equality comes from an application of the matrix inversion lemma. Substituting this expression into Equation (9.83) gives

$$P_m = P_{fm}^+ - K_m(P_{f,m+1}^- - P_{m+1})K_m^T \tag{9.90}$$

where the smoother gain K_m is given as

$$K_m = P_{fm}^+F_m^T\mathcal{I}_{f,m+1}^- \tag{9.91}$$

The covariance update equation for P_m is not a function of the backward covariance. The smoother covariance P_m can be solved by using only the forward covariance P_{fm}, which reduces the computational effort (compared to the algorithm presented in Section 9.4.1).

9.4.2.2 RTS state estimate update Next we consider the smoothed estimate $\hat{x}_m$ given in Equation (9.78). We will find an equivalent expression that does not use P_{bm} or $\hat{x}_{bm}$. In order to do this we will first need to establish a few lemmas.

Lemma 1

$$F_{k-1}^{-1}Q_{k-1}F_{k-1}^{-T} = F_{k-1}^{-1}P_{fk}^{-}F_{k-1}^{-T} - P_{f,k-1}^{+} \tag{9.92}$$

Proof: From Equation (9.67) we see that

$$P_{fk}^{-} = F_{k-1}P_{f,k-1}^{+}F_{k-1}^{T} + Q_{k-1} \tag{9.93}$$

Rearranging this equation gives

$$Q_{k-1} = P_{fk}^{-} - F_{k-1}P_{f,k-1}^{+}F_{k-1}^{T} \tag{9.94}$$

Premultiplying both sides by F_{k-1}^{-1} and postmultiplying both sides by F_{k-1}^{-T} gives the desired result.
QED

Lemma 2 *The a posteriori covariance P_{bk}^{+} of the backward filter satisfies the equation*

$$P_{bk}^{+} = (P_{fk}^{-} + P_{bk}^{+})\mathcal{I}_{fk}^{-}P_k \tag{9.95}$$

Proof: From Equation (9.78) we obtain

$$\begin{aligned} I &= (\mathcal{I}_{bk}^{+} + \mathcal{I}_{fk}^{-})P_k \\ P_{bk}^{+} &= (I + P_{bk}^{+}\mathcal{I}_{fk}^{-})P_k \\ &= P_k + P_{bk}^{+}\mathcal{I}_{fk}^{-}P_k \\ &= P_{fk}^{-}\mathcal{I}_{fk}^{-}P_k + P_{bk}^{+}\mathcal{I}_{fk}^{-}P_k \\ &= (P_{fk}^{-} + P_{bk}^{+})\mathcal{I}_{fk}^{-}P_k \end{aligned} \tag{9.96}$$

QED

Lemma 3 *The covariances of the forward and backward filters satisfy the equation*

$$P_{fk}^{-} + P_{bk}^{+} = F_{k-1}(P_{f,k-1}^{+} + P_{b,k-1}^{-})F_{k-1}^{T} \tag{9.97}$$

Proof: From Equation (9.67) and (9.72) we see that

$$\begin{aligned} P_{f,k-1}^{+} &= F_{k-1}^{-1}P_{fk}^{-}F_{k-1}^{-T} - F_{k-1}Q_{k-1}F_{k-1}^{-T} \\ P_{b,k-1}^{-} &= F_{k-1}^{-1}P_{bk}^{+}F_{k-1}^{-T} + F_{k-1}^{-1}Q_{k-1}F_{k-1}^{-T} \end{aligned} \tag{9.98}$$

Adding these two equations and rearranging gives

$$\begin{aligned} P_{f,k-1}^{+} + P_{b,k-1}^{-} &= F_{k-1}^{-1}(P_{fk}^{-} + P_{bk}^{+})F_{k-1}^{-T} \\ P_{fk}^{-} + P_{bk}^{+} &= F_{k-1}(P_{f,k-1}^{+} + P_{b,k-1}^{-})F_{k-1}^{T} \end{aligned} \tag{9.99}$$

QED

Lemma 4 *The smoothed estimate $\hat{x}_k$ can be written as*

$$\hat{x}_k = P_k\mathcal{I}_{fk}^{+}\hat{x}_{fk}^{-} - P_kH_k^TR_k^{-1}H_k\hat{x}_{fk}^{-} + P_ks_k^{+} \tag{9.100}$$

Proof: From Equations (9.69) and (9.82) we have

$$\begin{aligned}\hat{x}_k &= P_k\mathcal{I}_{fk}^+\hat{x}_{fk}^+ + P_k\mathcal{I}_{bk}^-\hat{x}_{bk}^- \\ &= P_k\mathcal{I}_{fk}^+\hat{x}_{fk}^+ + P_k s_k^- \end{aligned} \tag{9.101}$$

From Equation (9.76) we see that

$$s_k^- = s_k^+ - H_k^T R_k^{-1} y_k \tag{9.102}$$

Substitute this expression for s_k^-, and the expression for $\hat{x}_{fk}^+$ from Equation (9.67), into Equation (9.101) to obtain

$$\hat{x}_k = P_k\mathcal{I}_{fk}^+\hat{x}_{fk}^- + P_k\mathcal{I}_{fk}^+K_{fk}(y_k - H_k\hat{x}_{fk}^-) + P_k s_k^+ - P_k H_k^T R_k^{-1} y_k \tag{9.103}$$

Now substitute $P_{fk}^+H_k^TR_k^{-1}$ for K_{fk} [from Equation (9.67)] in the above equation to obtain

$$\begin{aligned}\hat{x}_k &= P_k\mathcal{I}_{fk}^+\hat{x}_{fk}^- + P_k\mathcal{I}_{fk}^+P_{fk}^+H_k^TR_k^{-1}(y_k - H_k\hat{x}_{fk}^-) + P_k s_k^+ - P_k H_k^T R_k^{-1} y_k \\ &= P_k\mathcal{I}_{fk}^+\hat{x}_{fk}^- + P_kH_k^TR_k^{-1}(y_k - H_k\hat{x}_{fk}^-) + P_k s_k^+ - P_k H_k^T R_k^{-1} y_k \\ &= P_k\mathcal{I}_{fk}^+\hat{x}_{fk}^- - P_kH_k^TR_k^{-1}H_k\hat{x}_{fk}^- + P_k s_k^+ \end{aligned} \tag{9.104}$$

QED

Lemma 5

$$(P_{f,k-1}^+ + P_{b,k-1}^-)^{-1} = F_{k-1}^T\mathcal{I}_{fk}^-(P_{fk}^- - P_k)\mathcal{I}_{fk}^-F_{k-1} \tag{9.105}$$

Proof: Recall from Equations (6.26) and (9.72) that

$$\begin{aligned}\mathcal{I}_{fk}^+ &= \mathcal{I}_{fk}^- + H_k^TR_k^{-1}H_k \\ \mathcal{I}_{bk}^+ &= \mathcal{I}_{bk}^- + H_k^TR_k^{-1}H_k \end{aligned} \tag{9.106}$$

Combining these two equations gives

$$\begin{aligned}\mathcal{I}_{bk}^+ &= \mathcal{I}_{bk}^- + \mathcal{I}_{fk}^+ - \mathcal{I}_{fk}^- \\ &= \left[\left(\mathcal{I}_{bk}^- + \mathcal{I}_{fk}^+\right)^{-1}\right]^{-1} - \mathcal{I}_{fk}^- \\ &= P_k^{-1} - \mathcal{I}_{fk}^- \\ P_{bk}^+ &= \left(\mathcal{I}_k - \mathcal{I}_{fk}^-\right)^{-1} \end{aligned} \tag{9.107}$$

where we have used Equation (9.78) in the above derivation. Substitute this expression for P_{bk}^+ into Equation (9.97) to obtain

$$\begin{aligned}F_{k-1}(P_{f,k-1}^+ + P_{b,k-1}^-)F_{k-1}^T &= P_{fk}^- + P_{bk}^+ \\ &= P_{fk}^- + \left(\mathcal{I}_k - \mathcal{I}_{fk}^-\right)^{-1} \end{aligned} \tag{9.108}$$

Invert both sides to obtain

$$
\begin{aligned}
F_{k-1}^{-T}(P_{f,k-1}^{+}+P_{b,k-1}^{-})^{-1}F_{k-1}^{-1} &= \left[P_{fk}^{-}+\left(\mathcal{I}_k-\mathcal{I}_{fk}^{-}\right)^{-1}\right]^{-1} \\
(P_{f,k-1}^{+}+P_{b,k-1}^{-})^{-1} &= F_{k-1}^{T}\left[P_{fk}^{-}+\left(\mathcal{I}_k-\mathcal{I}_{fk}^{-}\right)^{-1}\right]^{-1}F_{k-1} \\
&= F_{k-1}^{T}\left[P_{fk}^{-}\mathcal{I}_{fk}^{-}P_{fk}^{-}+P_{fk}^{-}\mathcal{I}_{fk}^{-}\left(\mathcal{I}_k-\mathcal{I}_{fk}^{-}\right)^{-1}\mathcal{I}_{fk}^{-}P_{fk}^{-}\right]^{-1}F_{k-1} \\
&= F_{k-1}^{T}\mathcal{I}_{fk}^{-}\left[\mathcal{I}_{fk}^{-}+\mathcal{I}_{fk}^{-}\left(\mathcal{I}_k-\mathcal{I}_{fk}^{-}\right)^{-1}\mathcal{I}_{fk}^{-}\right]^{-1}\mathcal{I}_{fk}^{-}F_{k-1} \qquad (9.109)
\end{aligned}
$$

Now apply the matrix inversion lemma to the term $(\mathcal{I}_k-\mathcal{I}_{fk}^{-})^{-1}$ in the above equation. This results in

$$
\begin{aligned}
&(P_{f,k-1}^{+}+P_{b,k-1}^{-})^{-1} \\
&= F_{k-1}^{T}\mathcal{I}_{fk}^{-}\left[\mathcal{I}_{fk}^{-}+\mathcal{I}_{fk}^{-}\left(-P_{fk}^{-}-P_{fk}^{-}(-P_{fk}^{-}+P_k)^{-1}P_{fk}^{-}\right)\mathcal{I}_{fk}^{-}\right]^{-1}\mathcal{I}_{fk}^{-}F_{k-1} \\
&= F_{k-1}^{T}\mathcal{I}_{fk}^{-}\left[\mathcal{I}_{fk}^{-}+\left(-I-(-P_{fk}^{-}+P_k)^{-1}P_{fk}^{-}\right)\mathcal{I}_{fk}^{-}\right]^{-1}\mathcal{I}_{fk}^{-}F_{k-1} \\
&= F_{k-1}^{T}\mathcal{I}_{fk}^{-}\left[\mathcal{I}_{fk}^{-}-\mathcal{I}_{fk}^{-}-(P_{fk}^{-}+P_k)^{-1}\right]^{-1}\mathcal{I}_{fk}^{-}F_{k-1} \\
&= F_{k-1}^{T}\mathcal{I}_{fk}^{-}(P_{fk}^{-}-P_k)\mathcal{I}_{fk}^{-}F_{k-1} \qquad (9.110)
\end{aligned}
$$

QED

With the above lemmas we now have the tools that we need to obtain an alternate expression for the smoothed estimate. Starting with the expression for s_{k-1}^{-} in Equation (9.76), and substituting the expression for $\mathcal{I}_{b,k-1}^{-}$ from Equation (9.72) gives

$$
\begin{aligned}
s_{k-1}^{-} &= \mathcal{I}_{bk}^{-}F_{k-1}^{-1}P_{bk}^{+}s_k^{+} \\
&= F_{k-1}^{T}\left[Q_{k-1}^{-1}-Q_{k-1}^{-1}\left(\mathcal{I}_{bk}^{+}+Q_{k-1}^{-1}\right)^{-1}Q_{k-1}^{-1}\right]F_{k-1}F_{k-1}^{-1}P_{bk}^{+}s_k^{+} \\
&= F_{k-1}^{T}Q_{k-1}^{-1}\left[I-\left(\mathcal{I}_{bk}^{+}+Q_{k-1}^{-1}\right)^{-1}Q_{k-1}^{-1}\right]P_{bk}^{+}s_k^{+} \\
&= F_{k-1}^{T}Q_{k-1}^{-1}\left(\mathcal{I}_{bk}^{+}+Q_{k-1}^{-1}\right)\left(\mathcal{I}_{bk}^{+}+Q_{k-1}^{-1}-Q_{k-1}^{-1}\right)P_{bk}^{+}s_k^{+} \\
&= F_{k-1}^{T}Q_{k-1}^{-1}\left(\mathcal{I}_{bk}^{+}+Q_{k-1}^{-1}\right)^{-1}s_k^{+} \\
&= F_{k-1}^{T}(I+\mathcal{I}_{bk}^{+}Q_{k-1})^{-1}s_k^{+} \qquad (9.111)
\end{aligned}
$$

Rearranging this equation gives

$$(I+\mathcal{I}_{bk}^{+}Q_{k-1})F_{k-1}^{-T}s_{k-1}^{-}=s_k^{+} \qquad (9.112)$$

Multiplying out this equation, and premultiplying both sides by $F_{k-1}^{-1}P_{bk}^{+}$, gives

$$F_{k-1}^{-1}P_{bk}^{+}F_{k-1}^{-T}s_{k-1}^{-}+F_{k-1}^{-1}Q_{k-1}F_{k-1}^{-T}s_{k-1}^{-}=F_{k-1}^{-1}P_{bk}^{+}s_k^{+} \qquad (9.113)$$

Substituting for $F_{k-1}^{-1}Q_{k-1}F_{k-1}^{-T}$ from Equation (9.92) gives

$$\begin{aligned}
F_{k-1}^{-1}P_{bk}^{+}F_{k-1}^{-T}s_{k-1}^{-} + F_{k-1}^{-1}P_{fk}^{-}F_{k-1}^{-T}s_{k-1}^{-} - P_{f,k-1}^{+}s_{k-1}^{-} &= F_{k-1}^{-1}P_{bk}^{+}s_{k}^{+} \\
\left[F_{k-1}^{-1}(P_{fk}^{-} + P_{bk}^{+})F_{k-1}^{-T} - P_{f,k-1}^{+}\right] s_{k-1}^{-} &= F_{k-1}^{-1}P_{bk}^{+}s_{k}^{+} \\
\left[(P_{fk}^{-} + P_{bk}^{+})F_{k-1}^{-T} - F_{k-1}P_{f,k-1}^{+}\right] s_{k-1}^{-} &= P_{bk}^{+}s_{k}^{+} \qquad (9.114)
\end{aligned}$$

Substituting in this expression for P_{bk}^{+} from Equation (9.95) gives

$$\left[(P_{fk}^{-} + P_{bk}^{+})F_{k-1}^{-T} - F_{k-1}P_{f,k-1}^{+}\right] s_{k-1}^{-} = (P_{fk}^{-} + P_{bk}^{+})\mathcal{I}_{fk}^{-}P_{k}s_{k}^{+} \qquad (9.115)$$

Substituting for $(P_{fk}^{-} + P_{bk}^{+})$ from Equation (9.97) on both sides of this expression gives

$$\left[F_{k-1}(P_{f,k-1}^{+} + P_{b,k-1}^{-}) - F_{k-1}P_{f,k-1}^{+}\right] s_{k-1}^{-} = F_{k-1}(P_{f,k-1}^{+} + P_{b,k-1}^{-})F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}s_{k}^{+} \qquad (9.116)$$

Premultiplying both sides by $(P_{f,k-1}^{+} + P_{b,k-1}^{-})^{-1}F_{k-1}^{-1}$ gives

$$\left[I - (P_{f,k-1}^{+} + P_{b,k-1}^{-})^{-1}P_{f,k-1}^{+}\right] s_{k-1}^{-} = F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}s_{k}^{+} \qquad (9.117)$$

Substituting Equation (9.105) for $(P_{f,k-1}^{+} + P_{b,k-1}^{-})^{-1}$ gives

$$s_{k-1}^{-} - F_{k-1}^{T}\mathcal{I}_{fk}^{-}(P_{fk}^{-} - P_{k})\mathcal{I}_{fk}^{-}F_{k-1}P_{f,k-1}^{+}s_{k-1}^{-} = F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}s_{k}^{+} \qquad (9.118)$$

Now from Equation (9.105) we see that

$$-(P_{f,k-1}^{+} + P_{b,k-1}^{-})^{-1}F_{k-1}^{-1}\hat{x}_{fk}^{-} = F_{k-1}^{T}\mathcal{I}_{fk}^{-}(P_{k} - P_{fk}^{-})\mathcal{I}_{fk}^{-}\hat{x}_{fk}^{-} \qquad (9.119)$$

So we can add the two sides of this equation to the two sides of Equation (9.118) to get

$$\begin{aligned}
&s_{k-1}^{-} - F_{k-1}^{T}\mathcal{I}_{fk}^{-}(P_{fk}^{-} - P_{k})\mathcal{I}_{fk}^{-}F_{k-1}P_{f,k-1}^{+}s_{k-1}^{-} - (P_{f,k-1}^{+} + P_{b,k-1}^{-})^{-1}F_{k-1}^{-1}\hat{x}_{fk}^{-} \\
&= F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}s_{k}^{+} + F_{k-1}^{T}\mathcal{I}_{fk}^{-}(P_{k} - P_{fk}^{-})\mathcal{I}_{fk}^{-}\hat{x}_{fk}^{-} \qquad (9.120)
\end{aligned}$$

Now use Equation (9.100) to substitute for $P_{k}s_{k}^{+}$ in the above equation and obtain

$$\begin{aligned}
&s_{k-1}^{-} - F_{k-1}^{T}\mathcal{I}_{fk}^{-}(P_{fk}^{-} - P_{k})\mathcal{I}_{fk}^{-}F_{k-1}P_{f,k-1}^{+}s_{k-1}^{-} - (P_{f,k-1}^{+} + P_{b,k-1}^{-})^{-1}F_{k-1}^{-1}\hat{x}_{fk}^{-} \\
&= F_{k-1}^{T}\mathcal{I}_{fk}^{-}\hat{x}_{k} - F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}\mathcal{I}_{fk}^{+}\hat{x}_{fk}^{-} + \\
&F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}H_{k}^{T}R_{k}^{-1}H_{k}\hat{x}_{fk}^{-} + F_{k-1}^{T}\mathcal{I}_{fk}^{-}(P_{k} - P_{fk}^{-})\mathcal{I}_{fk}^{-}\hat{x}_{fk}^{-} \qquad (9.121)
\end{aligned}$$

Rearrange this equation to obtain

$$\begin{aligned}
&s_{k-1}^{-} - F_{k-1}^{T}\mathcal{I}_{fk}^{-}(P_{fk}^{-} - P_{k})\mathcal{I}_{fk}^{-}F_{k-1}P_{f,k-1}^{+}s_{k-1}^{-} + \\
&\left[-(P_{f,k-1}^{+} + P_{b,k-1}^{-})^{-1}F_{k-1}^{-1} + F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}\mathcal{I}_{fk}^{+} - F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}H_{k}^{T}R_{k}^{-1}H_{k} - \right. \\
&\left. F_{k-1}^{T}\mathcal{I}_{fk}^{-}P_{k}\mathcal{I}_{fk}^{-}\right]\hat{x}_{fk}^{-} = F_{k-1}^{T}\mathcal{I}_{fk}^{-}(\hat{x}_{k} - \hat{x}_{fk}^{-}) \qquad (9.122)
\end{aligned}$$

From Equation (9.106) we see that $\mathcal{I}_{fk}^{+} - \mathcal{I}_{fk}^{-} = H_k^T R_k^{-1} H_k$. Also note that part of the coefficient of $\hat{x}_{fk}^{-}$ on the left side of the above equation can be expressed as

$$(P_{f,k-1}^{+} + P_{b,k-1}^{-})^{-1} F_{k-1}^{-1} = \mathcal{I}_{b,k-1}^{-}(I + P_{f,k-1}^{+}\mathcal{I}_{b,k-1}^{-})^{-1} F_{k-1}^{-1} \tag{9.123}$$

From Equation (9.67) we see that $F_{k-1}^{-1}\hat{x}_{fk}^{-} = \hat{x}_{f,k-1}^{+}$. Therefore Equation (9.122) can be written as

$$\begin{aligned} & s_{k-1}^{-} - F_{k-1}^T \mathcal{I}_{fk}^{-}(P_{fk}^{-} - P_k)\mathcal{I}_{fk}^{-} F_{k-1} P_{f,k-1}^{+} s_{k-1}^{-} - \\ & \mathcal{I}_{b,k-1}^{-}(I + P_{f,k-1}^{+}\mathcal{I}_{b,k-1}^{-})^{-1}\hat{x}_{f,k-1}^{+} = F_{k-1}^T \mathcal{I}_{fk}^{-}(\hat{x}_k - \hat{x}_{fk}^{-}) \end{aligned} \tag{9.124}$$

Now substitute for P_k from Equation (9.90) and use Equation (9.91) in the above equation to obtain

$$\begin{aligned} & s_{k-1}^{-} - F_{k-1}^T \mathcal{I}_{fk}^{-}(P_{fk}^{-} - P_{fk}^{+} + K_k P_{f,k+1}^{-} K_k^T - K_k P_{k+1} K_k^T) K_{k-1}^T s_{k-1}^{-} - \\ & \mathcal{I}_{b,k-1}^{-}(I + P_{f,k-1}^{+}\mathcal{I}_{b,k-1}^{-})^{-1}\hat{x}_{f,k-1}^{+} = F_{k-1}^T \mathcal{I}_{fk}^{-}(\hat{x}_k - \hat{x}_{fk}^{-}) \end{aligned} \tag{9.125}$$

Premultiplying both sides by $P_{f,k-1}^{+}$ gives

$$\begin{aligned} & \left[P_{f,k-1}^{+} - P_{f,k-1}^{+} F_{k-1}^T \mathcal{I}_{fk}^{-}(P_{fk}^{-} - P_{fk}^{+} + K_k P_{f,k+1}^{-} K_k^T - K_k P_{k+1} K_k^T) K_{k-1}^T\right] s_{k-1}^{-} \\ & -P_{f,k-1}^{+}\mathcal{I}_{b,k-1}^{-}(I + P_{f,k-1}^{+}\mathcal{I}_{b,k-1}^{-})^{-1}\hat{x}_{f,k-1}^{+} = K_{k-1}(\hat{x}_k - \hat{x}_{fk}^{-}) \end{aligned} \tag{9.126}$$

Now use Equation (9.91) to notice that the coefficient of s_{k-1}^{-} on the left side of the above equation can be written as

$$P_{f,k-1}^{+} - K_{k-1}\left(P_{fk}^{-} - P_{fk}^{+} + K_k P_{f,k+1}^{-} K_k^T - K_k P_{k+1} K_k^T\right) K_{k-1}^T \tag{9.127}$$

Using Equation (9.90) to substitute for $K_k P_{k+1} K_k^T$ allows us to write the above expression as

$$\begin{aligned} & P_{f,k-1}^{+} - K_{k-1}\left(P_{fk}^{-} - P_{fk}^{+} + K_k P_{f,k+1}^{-} K_k^T - P_k + P_{fk}^{+} - K_k P_{f,k+1}^{-} K_k^T\right) K_{k-1}^T \\ & \quad = P_{f,k-1}^{+} - K_{k-1} P_{fk}^{-} K_{k-1}^T + K_{k-1} P_k K_{k-1}^T \\ & \quad = P_{f,k-1}^{+} - K_{k-1} P_{fk}^{-} K_{k-1}^T + P_{k-1} - P_{f,k-1}^{+} + K_{k-1} P_{fk}^{-} K_{k-1}^T \\ & \quad = P_{k-1} \end{aligned} \tag{9.128}$$

Since this is the coefficient of s_{k-1}^{-} in Equation (9.126), we can write that equation as

$$P_{k-1} s_{k-1}^{-} - P_{f,k-1}^{+}\mathcal{I}_{b,k-1}^{-}(I + P_{f,k-1}^{+}\mathcal{I}_{b,k-1}^{-})^{-1}\hat{x}_{f,k-1}^{+} = K_{k-1}(\hat{x}_k - \hat{x}_{fk}^{-}) \tag{9.129}$$

Now from Equations (9.78) and (9.82) we see that

$$\begin{aligned} \hat{x}_k &= (\mathcal{I}_{fk}^{+} + \mathcal{I}_{bk}^{-})^{-1}\mathcal{I}_{fk}^{+}\hat{x}_{fk}^{+} + P_k \mathcal{I}_{bk}^{-}\hat{x}_{bk}^{-} \\ &= (I + P_{fk}^{+}\mathcal{I}_{bk}^{-})^{-1}\hat{x}_{fk}^{+} + P_k s_k^{-} \end{aligned} \tag{9.130}$$

From this we see that

$$\begin{aligned} \hat{x}_k - \hat{x}_{fk}^{+} &= \left[(I + P_{fk}^{+}\mathcal{I}_{bk}^{-})^{-1} - I\right]\hat{x}_{fk}^{+} + P_k s_k^{-} \\ &= \left[I - (I + P_{fk}^{+}\mathcal{I}_{bk}^{-})\right](I + P_{fk}^{+}\mathcal{I}_{bk}^{-})^{-1}\hat{x}_{fk}^{+} + P_k s_k^{-} \\ &= -P_{fk}^{+}\mathcal{I}_{bk}^{-}(I + P_{fk}^{+}\mathcal{I}_{bk}^{-})^{-1}\hat{x}_{fk}^{+} + P_k s_k^{-} \end{aligned} \tag{9.131}$$

Rewriting the above equation with the time subscripts $(k-1)$ and then substituting for the left side of Equation (9.129) gives

$$\hat{x}_{k-1} - \hat{x}^+_{f,k-1} = K_{k-1}(\hat{x}_k - \hat{x}^-_{fk}) \tag{9.132}$$

from which we can write

$$\hat{x}_k = \hat{x}^+_{fk} + K_k(\hat{x}_{k+1} - \hat{x}^-_{f,k+1}) \tag{9.133}$$

This gives the smoothed estimate $\hat{x}_k$ without needing to explicitly calculate the backward estimate. The RTS smoother is implemented by first running the standard Kalman filter of Equation (9.67) forward in time to the final time, and then implementing Equations (9.90), (9.91), and (9.133) backward in time. The RTS smoother can be summarized as follows.

The RTS smoother

1. The system model is given as follows:

$$\begin{aligned} x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned} \tag{9.134}$$

2. Initialize the forward filter as follows:

$$\begin{aligned} \hat{x}_{f0} &= E(x_0) \\ P^+_{f0} &= E\left[(x_0 - \hat{x}_{f0})(x_0 - \hat{x}_{f0})^T\right] \end{aligned} \tag{9.135}$$

3. For $k = 1, \cdots, N$ (where N is the final time), execute the standard forward Kalman filter:

$$\begin{aligned} P^-_{fk} &= F_{k-1}P^+_{f,k-1}F^T_{k-1} + Q_{k-1} \\ K_{fk} &= P^-_{fk}H^T_k(H_kP^-_{fk}H^T_k + R_k)^{-1} \\ &= P^+_{fk}H^T_kR^{-1}_k \\ \hat{x}^-_{fk} &= F_{k-1}\hat{x}^+_{f,k-1} + G_{k-1}u_{k-1} \\ \hat{x}^+_{fk} &= \hat{x}^-_{fk} + K_{fk}(y_k - H_k\hat{x}^-_{fk}) \\ P^+_{fk} &= (I - K_{fk}H_k)P^-_{fk}(I - K_{fk}H_k)^T + K_{fk}R_kK^T_{fk} \\ &= \left[(P^-_{fk})^{-1} + H^T_kR^{-1}_kH_k\right]^{-1} \\ &= (I - K_{fk}H_k)P^-_{fk} \end{aligned} \tag{9.136}$$

4. Initialize the RTS smoother as follows:

$$\begin{aligned} \hat{x}_N &= \hat{x}^+_{fN} \\ P_N &= P^+_{fN} \end{aligned} \tag{9.137}$$